

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.



⚠ WARNING ⚠

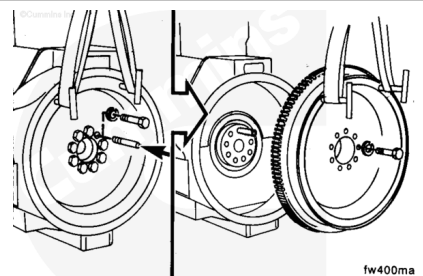
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries or air starter. Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html).
- Remove the transmission, clutch, and all related components. Refer to the manufacturer's operating instructions.

Remove

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This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use two [5/8-18 inch] guide studs to prevent the flywheel from rotating. Remove two capscrews and install the guide studs.

Use a hoist, two tee handles, and a lifting sling. Install the tee handles.

Remove the remaining capscrews.

Remove the flywheel.

Note : Use a mallet to tap the flywheel from the crankshaft, if necessary.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

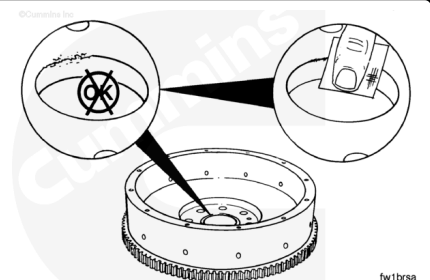
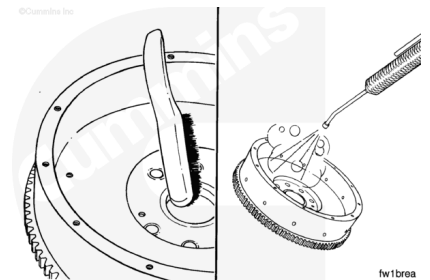
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a wire brush to clean the crankshaft pilot bore. Use steam or solvent to clean the flywheel. Dry with compressed air.

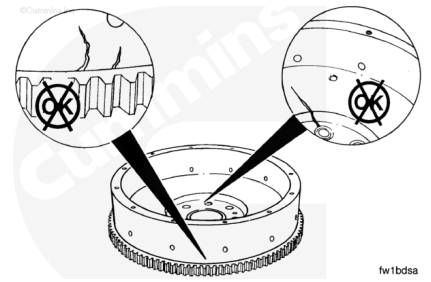
Inspect for nicks or burrs. Use a crocus cloth to remove small nicks and burrs.



⚠ WARNING ⚠

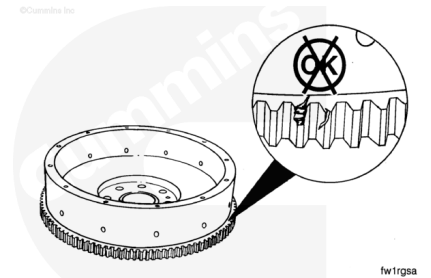
Do not use a cracked flywheel. A cracked flywheel can break and cause serious personal injury.

Inspect the flywheel for cracks.



Inspect the flywheel ring gear teeth for cracks and chips.

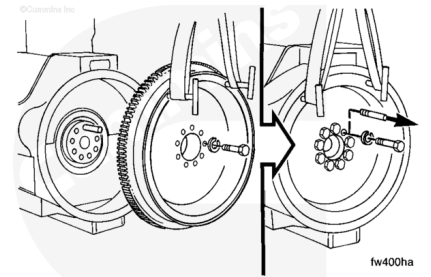
Note : If the ring gear teeth are cracked or broken, the ring gear must be replaced.



Install

⚠ WARNING ⚠

The flywheel mounting capscrews must be a minimum of SAE Grade 8 with rolled threads. The flywheel mounting washers are special hardened plain washers. Use identical replacements to reduce possible flywheel failure resulting in personal injury or property damage.



⚠ WARNING ⚠

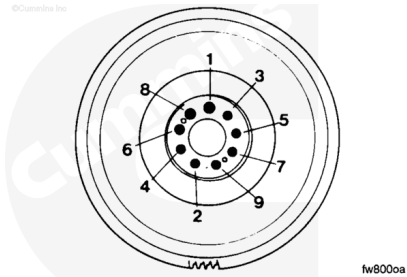
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Lubricate the capscrews and washers with engine oil. Allow the excess oil to drip off the parts. Do **not** lubricate the threads of the crankshaft.

A guide stud will help during assembly. Install the flywheel, washers, and capscrews. The flywheel **must** be, firmly seated against the crankshaft.

Use the following steps and the torque sequence shown to install the flywheel.

Torque Value:	Step 1	150 n.m [110 ft-lb]
	Step 2	285 n.m [210 ft-lb]



Measure the alignment of the flywheel.

Note : The crankshaft end clearance must be pushed or pulled in the same direction each time a point is measured.

Attach an indicator as shown. Measure the flywheel alignment at 4 equally spaced points.

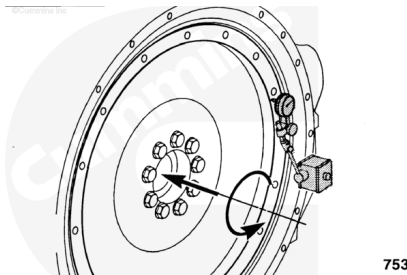
Measure the distance from the center of the crankshaft to the indicator tip. Multiply the distance to obtain the maximum runout.

The alignment specification is 0.025 mm [0.001 inch] per 25.4 mm [1.0 inch] distance from the center of the crankshaft.

Center of Crankshaft-To-Indicator Tip:

Multiply by 0.025 mm [0.001 inch]

If the flywheel alignment is **not** within specifications, check for interference between the flywheel and the crankshaft.

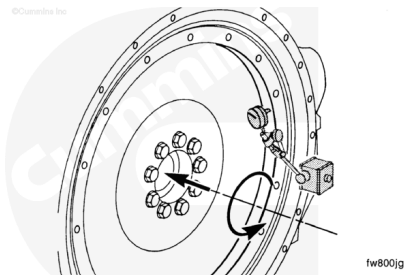


Attach an indicator as shown. Continue to look at the indicator while rotating the engine.

Measurements		
	mm	in
Maximum Radial Runout	0.13	0.005

If the runout is **not** within specification, the pilot on the flywheel is **not** positioned correctly on the crankshaft.

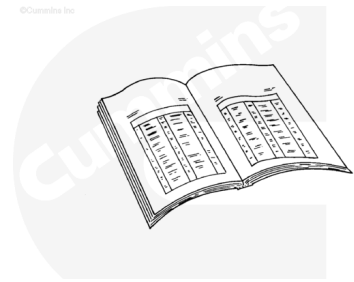
If the pilot is damaged, the flywheel **must** be replaced.



Finishing Steps

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- Install the transmission, clutch, and all related components. Refer to the equipment manufacturer's operating instructions.
- Connect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Start the engine. Check for proper operation and leaks.